MTH1002-A

UNIVERSITY OF EXETER

FACULTY OF ENVIRONMENT, SCIENCE AND ECONOMY

MATHEMATICS AND STATISTICS

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Mathematical Methods

Module Leader: Dr Layal Hakim

Duration: 2 HOURS

Answer Section A (50%) and any TWO of the three questions in Section B (25% for each).

Marks shown in questions are merely a guideline. Candidates are permitted to use approved portable electronic calculators in this examination.

This is a **CLOSED NOTE** examination.

SECTION A

1. (a) Consider the cubic equation

$$y = x^3 + k.$$

Find the values of k for which the line $y = 3x$ is tangent to the cubic. (10)

(b) Apply De Moivre's Theorem to solve the equation $z^4 = 1$, thus obtaining the fourth roots of 1. (10)

(c) Let f and g be functions with the following rules: $f(x) = e^x$, and $g(x) = \sin(x)$. Find $f \circ g$, and determine whether this composition is (i) even, (ii) odd, and (iii) periodic. (10)

(d) Given the three points $A = (0, 0, 0)$, $B = (1, -3, 1)$, and $C = (3, 1, 1)$, find the angle between $\overrightarrow{AB}$ and $\overrightarrow{AC}$. (10)

(e) Write the following series in the form $\sum_n a_n$, and use this to decide whether or not the series converges:

$$\frac{1}{5} + \frac{2}{6} + \frac{2^2}{7} + \frac{2^3}{8} + \frac{2^4}{9} + \cdots .$$

(10)

[**50**]

SECTION B

2. (a) Let

$$f(x) = \frac{1-x}{2+6x+x^2}.$$

(i) By finding the asymptotes and stationary points, sketch $f(x)$. (9)

(ii) Give the largest subset of the real numbers for which $f(x)$ defines a function. (1)

(iii) Give a subset of the real numbers on which f is monotonic increasing. (1)

(iv) Give a subset of the real numbers on which f is monotonic decreasing. (1)

(b) Determine whether the following series is convergent or divergent:

$$\sum_{n=1}^{\infty} \frac{\cos(n^5 - 2n)}{n^2}.$$

(5)

(c) Show from first principles, that is, using the formal definition of a derivative, that the derivative of $f(x) = \sqrt{x}$ is $f'(x) = \frac{1}{2\sqrt{x}}$. (8)

[25]

3. (a) Write the following polar equation in Cartesian form. Hence, or otherwise, describe the conic section this equation represents.

$$r = 6\cos\theta - 8\sin\theta$$

(6)

(b) Compute the following limit:

$$\lim_{x\to 0} \frac{e^{4x} - 1 - 4x - 8x^2}{x^3}.$$

(6)

(c) (i) Show that the Maclaurin series expansion for $f(x) = \frac{1}{1-x}$ is $f(x) = 1 + x + x^2 + x^3 + x^4 + \cdots$. (5)

(ii) Write $f(x)$ in Σ format, i.e. $f(x) = \sum_{k=0}^{\infty} a_k x^k$. Find the radius of convergence for this power series. (8)

[25]

4. (a) (i) State Taylor's theorem for the expansion of a function $f(x)$ about the point $x = 2$, up to the term in $(x-2)^4$. (3)

(ii) Find the Taylor series expansion of $f(x) = x\ln(x)$ about the point $x = 2$ up to terms of order 4. (6)

(b) Using De Moivre's Theorem or otherwise, show that

$$\sin^3\theta = -\frac{1}{4}\sin(3\theta) + \frac{3}{4}\sin(\theta).$$

(8)

(c) Three points are given by their Cartesian coordinates $A = (1,1,1), B = (2,1,-3)$ and $C = (-2,2,1)$.

(i) Provide the coordinates of the normal vector to the plane which passes through the points A, B and C. (4)

(ii) Find the equation of the plane which passes through the points A, B and C in the Cartesian form $\alpha x + \beta y + \gamma z + \delta = 0$. (4)

[**25**]